

1) ~~ADD B~~
 $A \leftarrow A + B$

2) ADD $\overline{m}$
 $A \leftarrow A + M$

3) ADI 25H
 $A \leftarrow A + 25H$

4) SUB B
 $A \leftarrow A - B$

5) SUB $\overline{m}$

6) SUI 25H

7) ~~ADD B~~
8) ADC $\overline{m}$

9) ACI 25H

10) SBB B

11) SBB $\overline{m}$

12) SBI 25H



2) ADD m
 $A \leftarrow A + m$

3) ADI 25H
 $A \leftarrow A + 25H$

4) SUB B
EF $A \leftarrow A - B$

5) SUB m
 $A \leftarrow A - m$

6) SUI 25H
 $A \leftarrow A - 25H$

8) ADC m

9) ACI 25H

10) SBB B

11) SBB m

12) SBI 25H

B
+B

7) ADC B

$$A \leftarrow A + B + CF$$

ADC (+) ADD

12	FF
00	01
13	00

8) ADC m

$$A \leftarrow A + m + CF$$

9) ACI 25H

$$A \leftarrow A + 25H + CF$$

10) SBB B

$$A \leftarrow A - B - CF$$

27
-19
08

11) SBB m

$$A \leftarrow A - m - CF$$

13) INR B

$$B \leftarrow B + 1$$

14) INX B

$$BC \leftarrow BC + 1$$

15) INR m

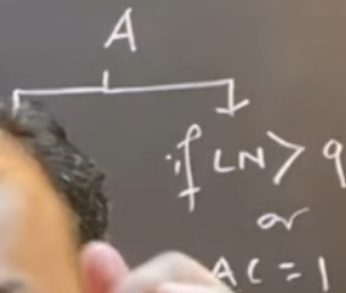
16) DCR B

17) DCR B

18) DCR

19) DAI

20) DAA {Decimal ad



13) INR B

$$B \leftarrow B + 1$$

14) INX B
 $BC \leftarrow BC + 1$

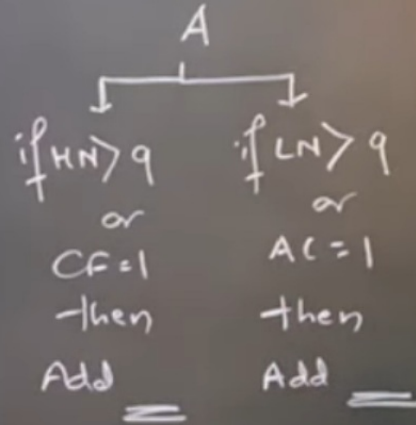
15) INR m
 $m \leftarrow m + 1$

16) DCR B
 $B \leftarrow B - 1$

17) DCR B
 $BC \leftarrow BC - 1$

18) DCR m
 $m \leftarrow m - 1$

19) DAD H
 $\underline{HL} \leftarrow \underline{HL} + \underline{HL}$



24H	20H	25H
+ 25H	+ 30H	+ 25H

40H	5
+ 50H	→

12) SBI 25H
 $A \leftarrow A - 25 - CF$

DAD - Double Addition

- Adds 16-bit numbers

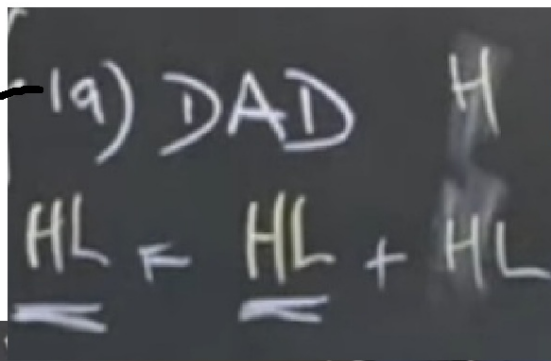
- DAD B $\Rightarrow$ HL HL + BC

- DAD D $\Rightarrow$ HL HL + DE

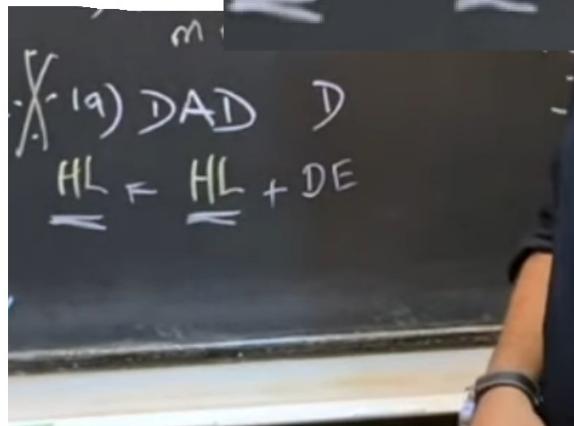
- DAD H $\Rightarrow$ HL HL + HL

- The last instr doubles the data of HL pair.

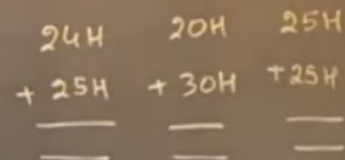
- If we have to multiply a number with 2 store it into HL & do DAD.



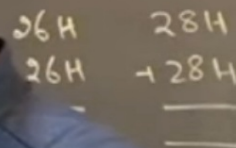
19) DAD H
HL $\leftarrow$ HL + HL



19) DAD D
HL $\leftarrow$ HL + DE



24H + 25H —	20H + 30H —	25H + 25H —
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26H + 26H —	28H + 28H —
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DAA (Decimal Adjustment after Addition)

A

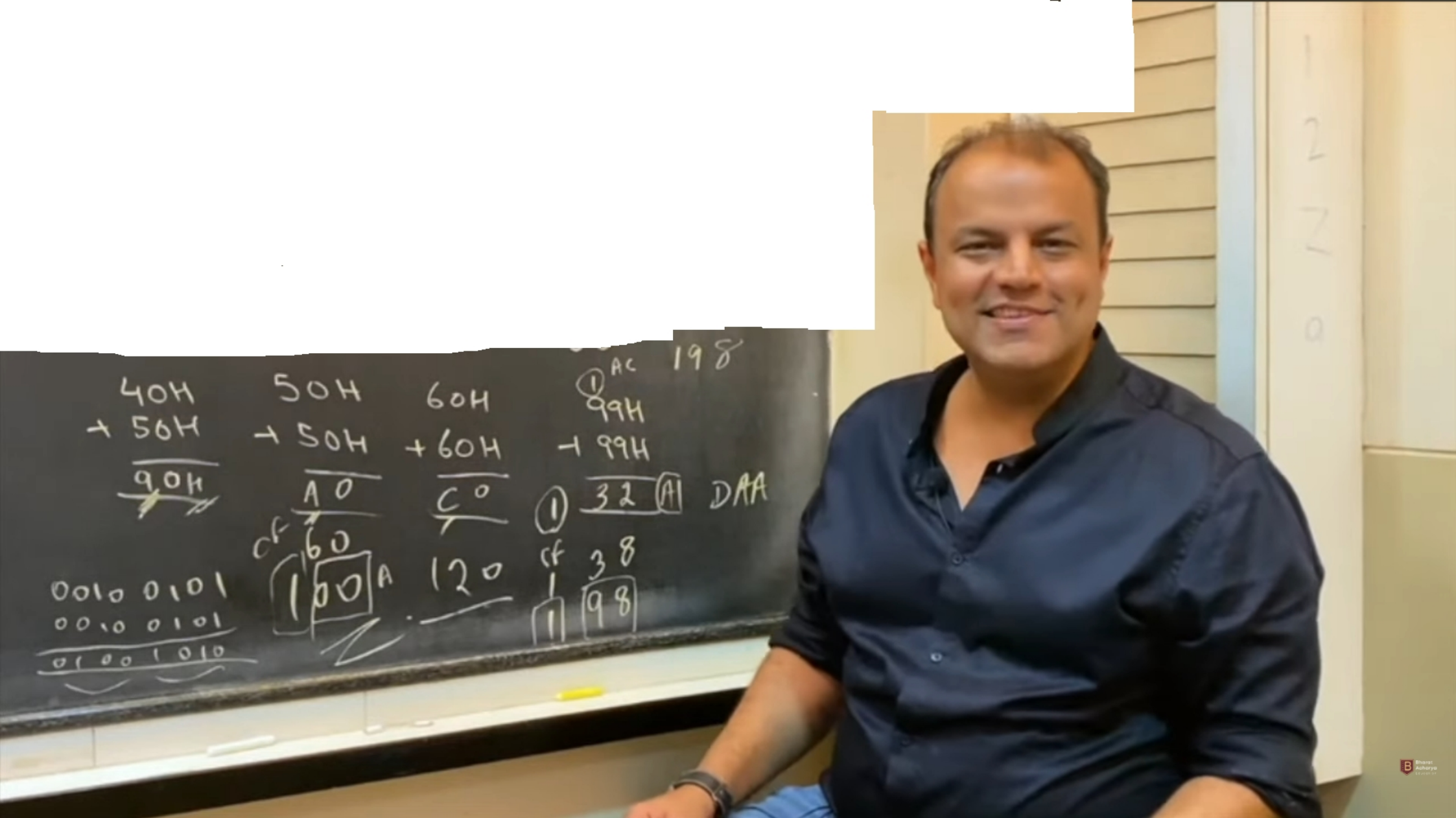
if $HN > 9$
or
 $CF = 1$
then
Add

if $LN > 9$
or
 $AC = 1$
then
Add 06

24H	20H	25H
+ 25H	+ 30H	+ 25H
M: <u>49</u>	<u>50</u>	<u>4A</u>
49	50	50

26H	28H
+ 26H	+ 28H
<u> </u>	<u> </u>
<u> </u>	<u> </u>

40H	50H	60H	99H
+ 50H	+ 50H	+ 60H	+ 99H
<u> </u>	<u> </u>	<u> </u>	<u> </u>
<u> </u>	<u> </u>	<u> </u>	<u> </u>



$$\begin{array}{r} 40H \\ + 50H \\ \hline 90H \\ \hline \end{array}$$

$$\begin{array}{r} 50H \\ + 50H \\ \hline A0 \\ \hline \end{array}$$

$$\begin{array}{r} 60H \\ + 60H \\ \hline C0 \\ \hline \end{array}$$

$$\begin{array}{r} \textcircled{1} \text{ AC } 198 \\ 99H \\ + 99H \\ \hline \textcircled{1} 32 \text{ (A)} \text{ DAA} \\ \hline \text{cf } 38 \\ \hline \text{11 } 98 \end{array}$$

$$\begin{array}{r} 0010 \ 0101 \\ 0010 \ 0101 \\ \hline 0100 \ 1010 \end{array}$$

$$\begin{array}{r} \text{cf } 60 \\ \hline \boxed{100} \text{ A} \\ \hline \end{array}$$

$$\begin{array}{r} 120 \\ \hline \end{array}$$